Analysis of existing heuristics and implementing a new obfuscation method for hiding influential individuals in community networks

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Introduction

- Network analysis used in various domains
- In social media, to explain
  - virality of content
  - polarization of opinions
- In pandemic modeling
  - identifying infection transmission pathways
  - forming effective vaccination strategies
- For national security purposes
  - analyzing terrorist networks
  - analyzing cyber attack pathways
- All social networks contain influential members.
- Network dynamics revolve around these “masterminds”
- Our focus - on social and crime networks

Problem Description and Motivation

- Identifying influential nodes is important
- Privacy concerns $\rightarrow$ Need for concealment of such nodes
- Privacy-utility tradeoff $\rightarrow$ Hiding while maintaining influence
- Optimization Problem

Key Motivations

- How terrorists hide ?
- "Global Government Requests Report" (Facebook's release)

Questions Addressed

- Can users disguise themselves in networks?
- Can we reduce detectability while maintaining influence?

Centrality Measures

Degree Centrality

number of neighbors

$$C_{degr}(G, v) = |N_G(v)|$$

Closeness Centrality

distance to all nodes

$$c_{clos}(G, v) = \sum_{w \in V \setminus \{v\}} \frac{1}{\lambda_G(v, w)}$$

Betweenness Centrality

control over shortest paths

$$c_{betw}(G, v) = \sum_{w,u \in V \setminus \{v\}} \frac{|\{p \in \pi_G(w, u) : v \in p\}|}{|\pi_G(w, u)|}$$

Objectives

First Objective

- Hide influential node by reducing centrality
- Recover lost influence via rewiring

Second Objective

- Hide community structure from detection algorithms

Rewiring & Modularity

Rewiring is restricted to the individual's immediate network neighborhood. Involves 2 actions :

- unfriending or disconnecting certain neighbours
- introducing certain neighbours to each other

Modularity $\rightarrow$ index for measuring the quality of any given com munity structure.

- High modularity denotes structures that have dense connections within communities and sparse connections between them.
- Community detection algorithms search for communities having high modularity.

Models of Influence

Linear Threshold Model

- Nodes activate based on threshold
- Influence depends on neighbors' weights

Independent Cascade Model

- Probabilistic activation
- One chance per edge

LT metric

We have the following setup for the linear threshold model :

- A node v has a random threshold $\theta_v \sim U[0,1]$
- A node v is influenced by each neighbour w according to a weight $b_{v,w}$ such that,

$$\sum_{w \text{ neighbor of } v} b_{v,w} \leq 1$$

- A node v becomes active when the sum of the weights from its active neighbours becomes greater than the internal threshold of node v i.e.

$$\sum_{w \text{ active neighbor of } v} b_{v,w} \geq \theta_v$$

IC metric

Given a graph $G = (V, E)$ and an active subset of nodes S, this model has the following characteristics :

- Each edge (v, w) has a probability p_{vw}
- If node v becomes active, it gets one chance to make w active with probability p_{vw}
- An active node gets a single chance to activate its direct neighbours. The activation stops when all the active nodes have exhausted their chances of activating their neighbours.
- A random number is generated from a $Unif(0, 1)$. If the random number is less than the probability value, the node gets activated otherwise it stays inactive.

Physical Interpretation of IC and LT metric

IC(mastermind)

- Average number of nodes influenced by the mastermind through probabilistic activation when cascade starts from mastermind. Captures the ease with which the mastermind's influence can probabilistically diffuse through the network.

LT(mastermind)

- Average number of nodes activated under threshold dynamics starting from the mastermind. It measures how far the influence goes.

LT Resistance

- Inverse of LT(mastermind). It measures how much the network prevents influence from spreading.

Hardness Information

- Disguising Degree Centrality $\rightarrow$ P problem
- Disguising Closeness & Betweenness Centrality $\rightarrow$ NP-complete
- Influence recovery $\rightarrow$ NP-complete
- Optimal solution is computationally hard

Influence Recovery

Optimisation Problem

$$\underset{A \subseteq (V \times V) \setminus (E \cup \hat{A}) : \forall_{v_i \in V} \, inf_{(V, E \cup A)}(v^\dagger, v_i) \geq f(v_i)}{\arg\min} |A|.$$

$inf(v^\dagger, vi) =$ influence of mastermind on node vi

$f(vi) =$ minimum influence mastermind should have on node vi

$A \subseteq (V \times V) \setminus (E \cup \hat{A}) =$ edges we are not allowed to add

$arg\ min\ |A| =$ find the smallest set of edge changes A such that the minimum influence is preserved.

Heuristics

- Provide near-optimal solutions
- Easy to implement
- Do not require full network knowledge

ROAM Heuristic

- **ROAM - Remove One Add Many**
  Given a budget b, the ROAM heuristic involves the following steps:
  Step 1: Remove the link between the source node("mastermind") i.e. $v\dagger$ and its neighbour of choice v_0.
  Step 2: Connect v_0 to $b-1$ nodes of choice, who are neighbours of $v\dagger$ but not of v_0 (if there are fewer than $b-1$ such neighbours, connect v_0 to all of them).

  Note: Choose v_0 to be the neighbour of $v\dagger$ with the most connections i.e. the 2nd most influential node in the network. v_0 is connected to the $b-1$ neighbours of $v\dagger$ with the least connections(i.e. the $v_i s$).

ROAM on 9/11 network

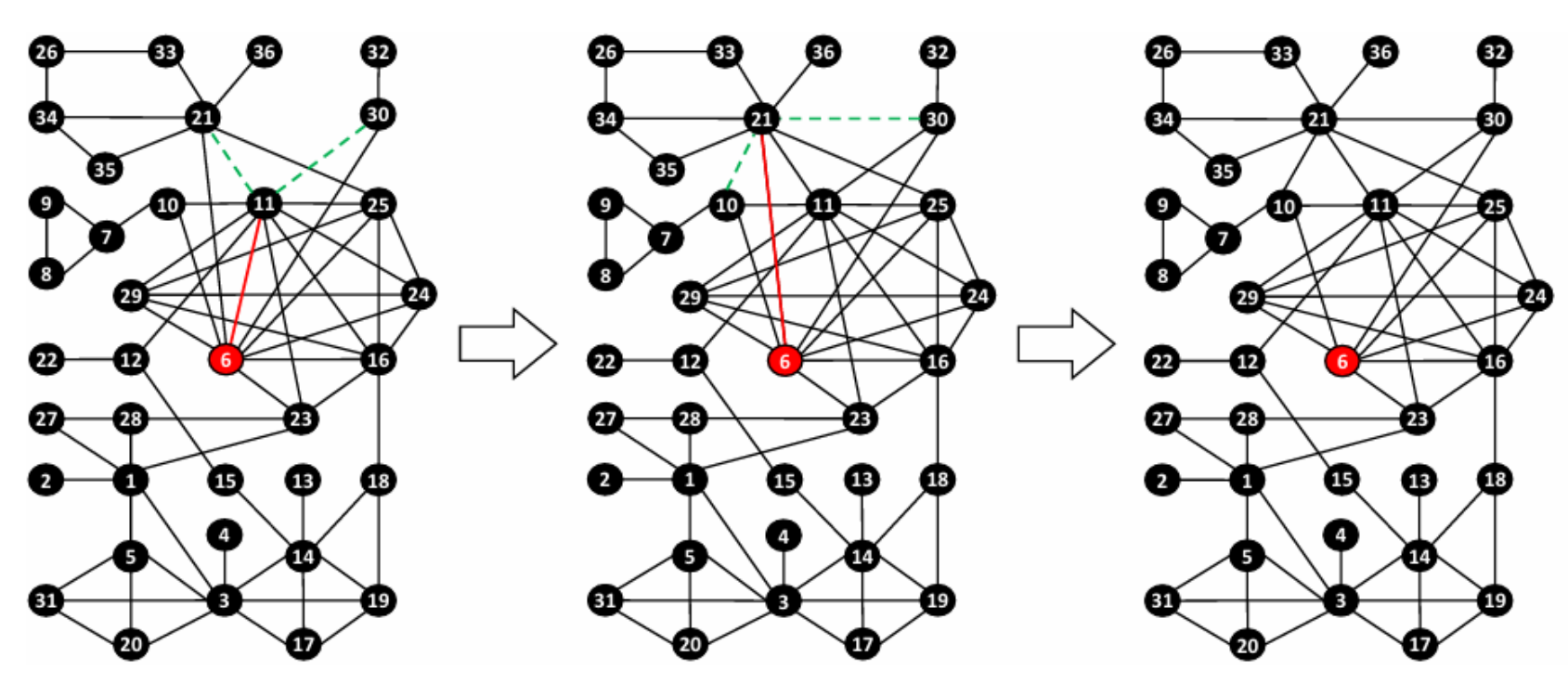


Figure 1: 9/11 terror attack network

ROAM simulation on 9/11 network

Metric	Initial	Final
Degree	0.257143	0.200000
Closeness	0.426829	0.380435
Betweenness	0.389380	0.109140
Degree Rank	1.000000	4.000000
Closeness Rank	1.000000	4.000000
Betweenness Rank	1.000000	11.000000
Sum of Ranks	3.000000	19.000000

Figure 2: Comparison of Initial and Final Centrality Measures of the mastermind node 6

Stage	IC Influence	LT Influence
Initial State	3.596	2.802
After Execution 1	3.252	2.972
After Execution 2	3.118	3.170

Figure 3: IC and LT Influence Values of the Mastermind Node

DICE Heuristic

- **DICE - Disconnect Internally, Connect Externally**

  Given a budget b, the **DICE heuristic** has the following steps:

  Step 1: Disconnect $d \leq b$ links from within $C^{\dagger}$.
  Step 2: Connect $b - d$ nodes from within $C^{\dagger}$ to $b - d$ nodes from outside of $C^{\dagger}$.
- Conceal a community $C^{\dagger}$
- Measure hiding by modularity

DICE simulation on 9/11 network

- Target community comprises nodes - (6, 11, 16, 23, 24, 25)
- The removed edges are: [(6, 16), (23, 16)]
- The added edges are : [(24, 19), (24, 13)]

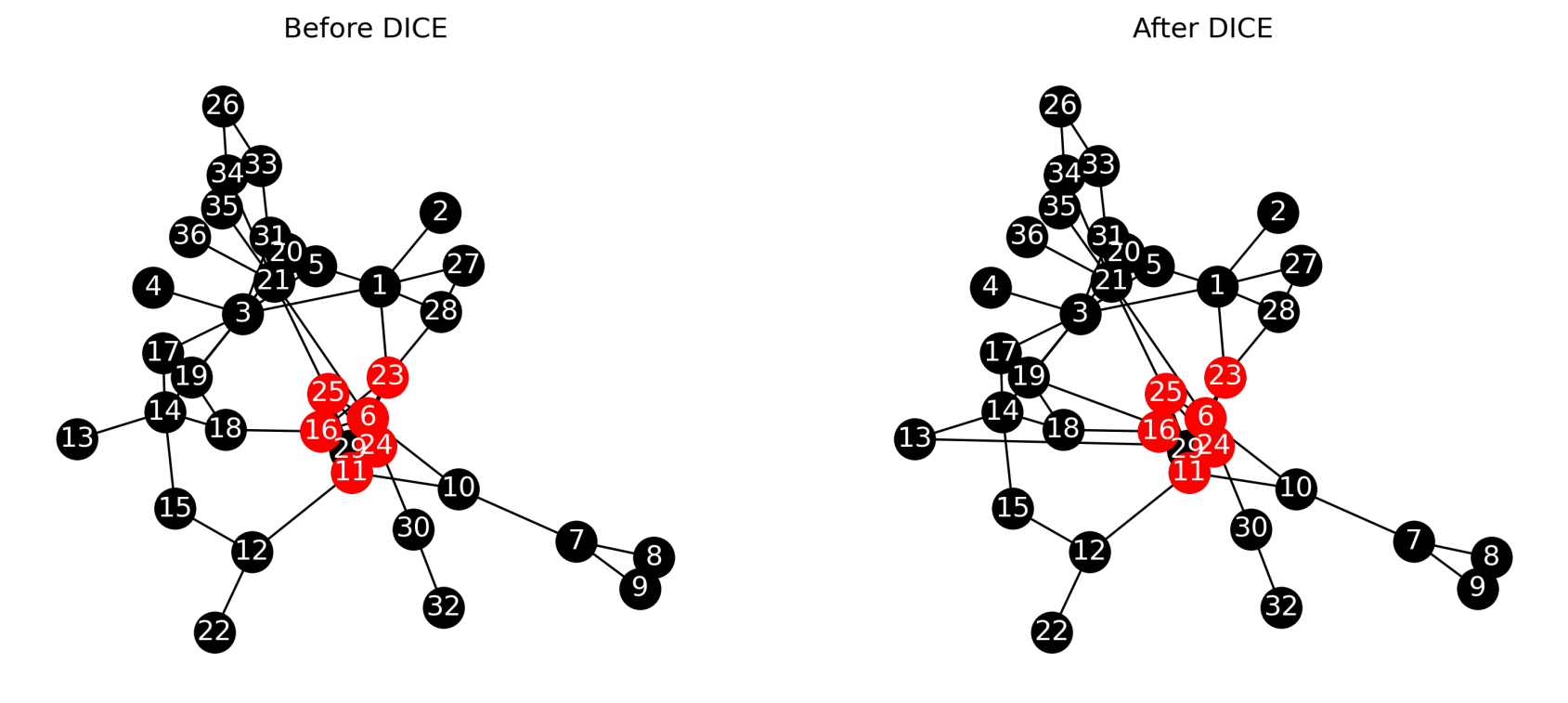


Figure 4: 9/11 terror network before and after DICE

DICE simulation on 9/11 network (continued)

Stage	Modularity
Before	0.2109375
After	0.16748046875

Figure 5: Modularity of $C^{\dagger}$ before and after execution of DICE

Modularity lesser $\rightarrow$ more sparse internally $\rightarrow$ harder to detect

HERMES Heuristic

- **HERMES - Hide Exposures by Removing Mastermind's External Sources**

1: Identify the core community C_i in the graph G.
2: Identify $M \in C_i$ based on the centrality measures.
3: Determine the set of external nodes ($NC^{\dagger}$) connected to M but not in C_i
4: **for** $k \leftarrow 1$ to $iter$ **do**
5: Find the top d most influential external nodes $NC_{max} \in NC^{\dagger}$ and remove the links between M and the nodes in NC_{max}. Reconnect these nodes with less influential nodes in C_i.
6: Remove these d external nodes from $NC^{\dagger}$.
7: **end for**

Figure 6: Algorithm

HERMES Heuristic (continued)

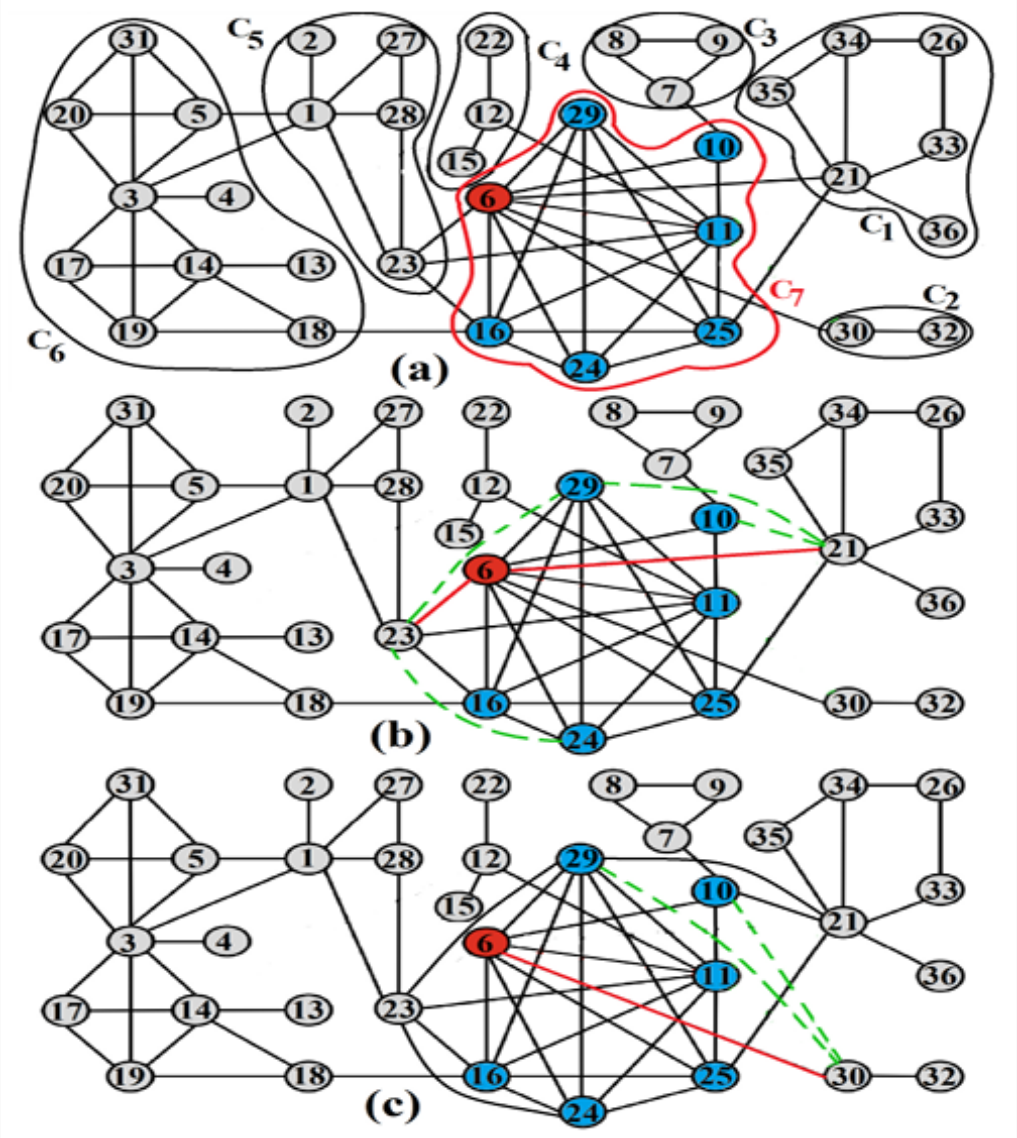


Figure 7: HERMES on 9/11 network

HERMES simulation on Facebook network dataset

- Original network $\rightarrow$ 4039 nodes and 88234 edges.
  Fragmented dataset consists of 333 nodes and 2978 edges.

Metric	Initial	Final
Degree Rank	95	101
Closeness Rank	81	203
Betweenness Rank	11	10
Sum of Ranks	187	314

Figure 8: Initial vs Final Ranks of Mastermind (Node 2724)

Stage	IC	LT
Initial	0.56	0.07
Iteration 1	0.54	0.08
Iteration 2	0.53	0.09

Figure 9: IC and LT values across HERMES iterations

C3R Heuristic

- **C3R - Community Constrained Cascade Re-routing**
- Community-based approach
- Multi-hop rewiring strategy
- Uses weakest nodes for re-routing influence

C3R Algorithm

- Partition the network into communities (e.g., Louvain).
- Rank nodes in each community using DC, CC, and BC.
- Identify the core community (lowest average rank).
- Select the mastermind as the most influential node in the core community.
- Identify external neighbors of the mastermind.
- Iteratively:
  - Select top d influential external nodes
  - Disconnect them from the mastermind
  - Rewire via weakest nodes:

$$O_i \rightarrow W_i \rightarrow MW_i \rightarrow \text{Mastermind}$$

C3R simulation on Facebook network dataset

Metric	Initial	Final
Degree Rank	91	93
Closeness Rank	89	333
Betweenness Rank	18	14
Sum of Ranks	198	440

Figure 10: Initial vs Final Centrality Ranks of Mastermind (Node 2724) for C3R

Stage	IC	LT
Initial	0.6304	0.007
Iteration 1	0.6264	0.010
Iteration 2	0.6089	0.011

Figure 11: IC and LT values across C3R iterations

IC, LT Comparison of HERMES and C3R

Stage	HERMES IC	HERMES LT	C3R IC	C3R LT
Initial	0.56	0.07	0.6304	0.007
Iteration 1	0.54	0.08	0.6264	0.010
Iteration 2	0.53	0.09	0.6089	0.011

Figure 12: Comparison of IC and LT Values for HERMES and C3R

Interpretation

- Value of IC at each iteration for HERMES is lower than that of C3R
- Value of LT at each iteration for HERMES is higher than that of C3R
- HERMES slightly outperforms our novel heuristic on the fragmented facebook dataset

Conclusion

- C3R needs testing over multiple networks to have an averaged assessment of its performance.
- In the IC model, each edge has a fixed probability $p = 0.15$. We can analyze the impact of varying activation probabilities in the IC model.
- In our modified LT model ($\delta = 0.05$), node thresholds increase after each failed activation, unlike the fixed-threshold standard LT model, making it more realistic for adversarial networks where nodes become progressively more resistant.
- Aside Louvain, explore alternative community formation algorithms like Girvan-Newman and Label Propagation.

Future Scope

- Develop improved heuristics for the hiding problem in complex networks.
- Investigate multilayer network scenarios where individuals exist across multiple platforms.
- Study related problems such as link prediction for forecasting future network connections.

Parting Note

Thank You !!

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